Enhancements to UseGalaxy.org Client-Facing Error Reporting for Optimized Functionality and Risk Mitigation

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1. Abstract

The Galaxy Project serves as a crucial platform for accessible, reproducible computational biology but, like any long-standing open source project, faces challenges that affect its efficiency. Imagine the scenario where dedicated researchers face an abrupt halt in their analytical tasks, triggered by system failures or cryptic error messages – occurring just as they reach critical milestones or moments before a pivotal meeting. These interruptions not only disrupt workflows but also often evoke a profound sense of helplessness and urgency. Although the Galaxy Project team strives to make user-facing errors infrequent, the occurrence of these issues underscores the ongoing need for enhancements in error handling and the development of more intuitive user interfaces to enable seamless scientific exploration.

Types of Errors on UseGalaxy.org

1. Data import/export
2. Tool execution
   a. **User-Reported Errors feature**
3. Workflow
4. Interface and usability
5. Performance
6. Authentication and security

Currently, when these types of errors occur, the user has a chance to self-report the error by submitting an error report through our online form which transmits an email, along with metadata according to a user's privacy settings, to the Galaxy-Bugs email list. After which, the Galaxy client support team may triage these incoming issues through a variety of means. In 2023, this engaged group initiated over 3,000 discussions via UseGalaxy.org, representing a dedicated minority who take extra steps to describe their challenges when encountering errors on the site. To address this subset of highly engaged users, this feature update focuses on significant improvements by consolidating client-facing self-reporting error processes, resolving data alignment issues, and improving the user experience during error encounters, all while leveraging existing tools for enhanced risk mitigation. By addressing these specific areas, our objective is to improve the reliability and efficiency of user-reported errors.

Improvements to User-Reported Errors feature

- Add extensibility to capture other error types
- Fix bugs, missing, misaligned metadata
- Component consolidation
- Automated reporting for risk management
- Improve integration with other user support features

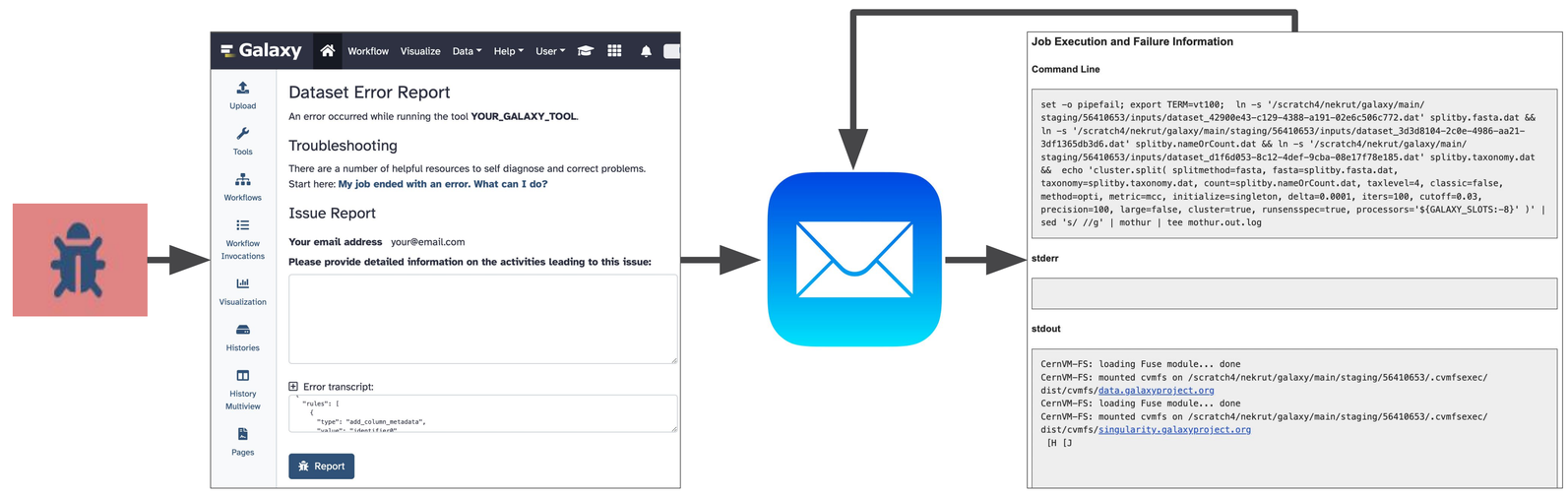
Footnotes

1 *User-reported error emails*
2 *Annual average per unique user 2014-2023*

Acknowledgements

We would like to thank Dr. Michael Schatz and the wonderful coworkers at the Galaxy Project for providing feedback on this work -especially, we would like to thank the original developers (past and present) having the insight and resourcefulness to build the first version of this feature. Major funding for the Galaxy Project is provided by NIH Grant 2U41HG006620. usegalaxy.org is supported by NIH and NSF Grants HG006620, 1661497, and 1929694. usegalaxy.eu is supported by the German Federal Ministry of Education and Research grant 031L0101C and de.NBI-epi. usegalaxy.org.au is supported by Bioplatforms Australia and the Australian Research Data Commons.

2. User-Reported Errors Feature

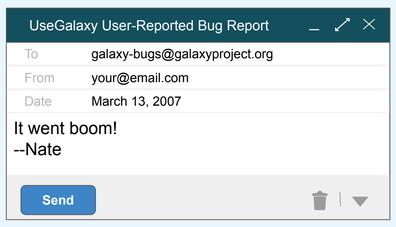


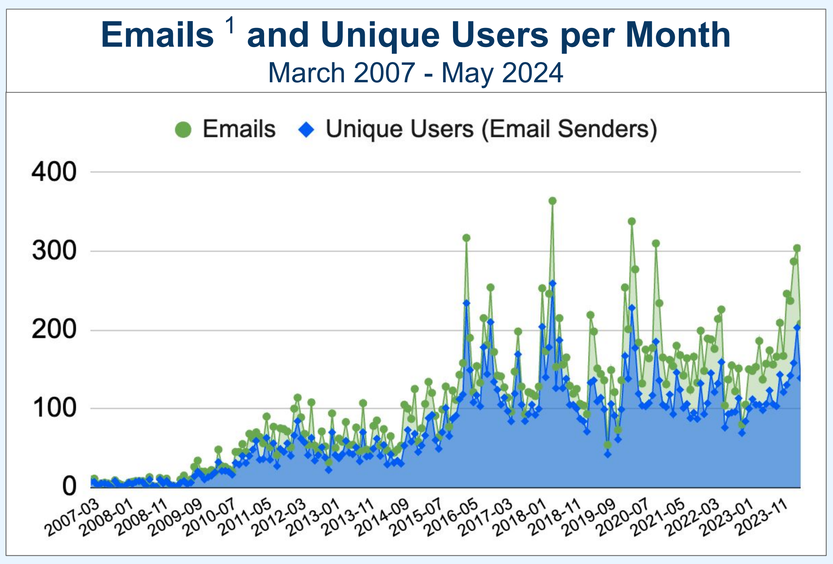
1. Error
2. User-Reporting Error Form *(ie. Dataset Error Report Page)*
3. Email Inboxes *(for US: Mailman Postgres server interface)*
4. Email contains User Message, Data Object Detail, Email Replies.

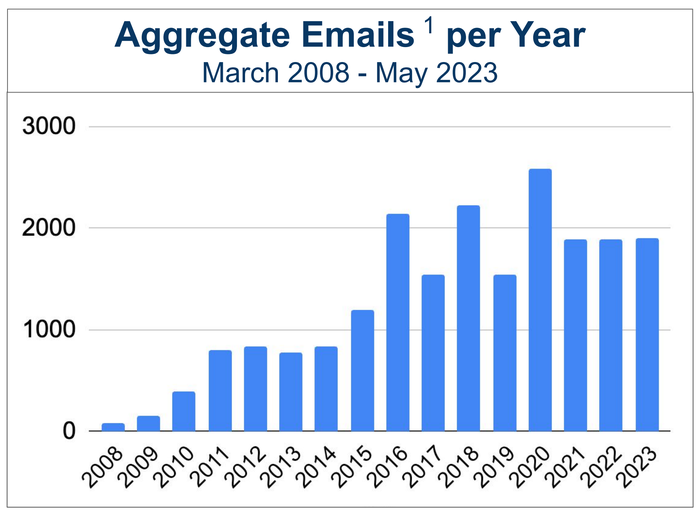
3. Key Metrics for User-Reported Error Emails

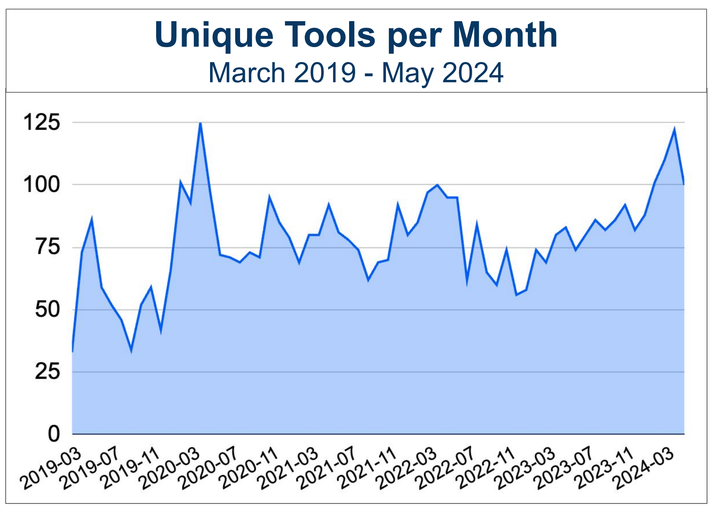
Key Metrics

- Total Emails [1] in 2023: **1,908 emails**
- Emails [1]: **1.4** per user [2]
- Date of First Email [1]: **March 14, 2007**



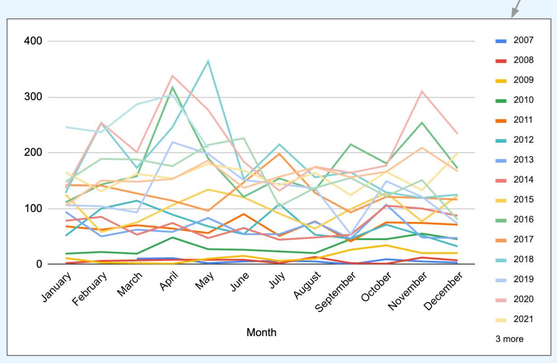


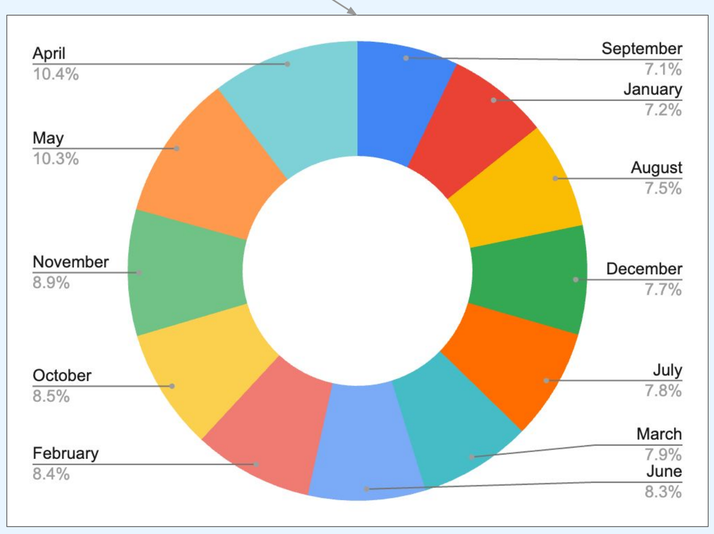




4. Initial Findings

The seasonal component highlights the **recurring patterns within each year**. It shows that certain months consistently have higher or lower message totals, confirming the **presence of seasonality**. For instance, months like April and May often see higher message totals.





ACTION ITEMS:

- Track & publish metrics after feature enhancements
- Explore integration of content in an automated system

What would *you* do with 17 years of email data?